Fundamentals of Statistical Learning

Definitions and results cheatsheet

Chapter 1. Defining probability

Definition 1.1: Outcomes and events

A random experiment is a process whose result is not known in advance. Each possible result is an *outcome* ω. The set of all possible outcomes is the *sample space* Ω. An *event* is a subset $A \subseteq \Omega$; it occurs when the observed outcome belongs to A.

Proposition 1.1: De Morgan's laws

For any events A and B,

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

More generally, for events $A_1, \dots, A_n$,

$$\left(\bigcup_{i=1}^{n} A_i\right)^c = \bigcap_{i=1}^{n} A_i^c, \qquad \left(\bigcap_{i=1}^{n} A_i\right)^c = \bigcup_{i=1}^{n} A_i^c.$$

Definition 1.2: Probability on a finite sample space

A probability measure P assigns a number $P(A)$ to every event $A \subseteq \Omega$ such that:

1. $0 \le P(A) \le 1$ for every event A;
2. $P(\Omega) = 1$;
3. if $A \cap B = \varnothing$, then $P(A \cup B) = P(A) + P(B)$.

The last property is called *finite additivity*.

Proposition 1.2: Inclusion–exclusion for two events

For any events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proposition 1.3: Union bound (Boole's inequality)

For events $A_1, \dots, A_n$,

$$P\left(\bigcup_{i=1}^{n} A_i\right) \le \sum_{i=1}^{n} P(A_i).$$

Proposition 1.4: Bonferroni's inequality

For events $A_1, \ldots, A_n$,

$$P\left(\bigcap_{i=1}^{n} A_i\right) \geq \sum_{i=1}^{n} P(A_i) - (n-1).$$

Chapter 2. Conditional probability

Definition 2.1: Conditional probability

For events A, B with $P(B) > 0$, the probability of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Definition 2.2: Independent events

Two events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

Theorem 2.1: Bayes' theorem

For a group B_j of the partition[†] and an event A with $P(A) > 0$,

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{P(A)} = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{m} P(A \mid B_i)P(B_i)}.$$

Definition 2.3: Prior and posterior

In Bayes' theorem, $P(B_j)$ is the probability of B_j before observing A, called the **prior**. Its update $P(B_j \mid A)$ after observing A is called the **posterior**.

Chapter 3. Random variables

Definition 3.1: Random object

A *random object* is a measurable[*] map from outcomes to possible observations:

$$Z : \Omega \longrightarrow \mathcal{T}, \qquad \omega \longmapsto Z(\omega).$$

The set $\mathcal{T}$ says what kind of thing we record. It could contain numbers, vectors, images, or text messages. Once ω is known, $Z(\omega)$ is the observation we get.

[†]Here $B_1, \ldots, B_m$ form a partition of the sample space, with $P(B_i) > 0$.

[*]A rigorous definition of measurability is beyond the scope of this course.

Definition 3.2: Random variable

A *random variable* is a measurable map

$$X : \Omega \longrightarrow \mathbb{R}.$$

It assigns one real number $X(\omega)$ to each outcome ω.

Definition 3.3: Random vector

A *p-dimensional random vector* is a measurable map

$$\underline{X} : \Omega \longrightarrow \mathbb{R}^p, \qquad \underline{X}(\omega) := \big(X_1(\omega), \ldots, X_p(\omega)\big).$$

Each coordinate X_j is a real-valued random variable. Thus one outcome gives us p numbers collected into a vector. These measurements can be related; they do not have to be independent.

Definition 3.4: Preimage

For a random variable $X : \Omega \to \mathbb{R}$ and a set $E \subseteq \mathbb{R}$, the *preimage* (or *inverse image*) of E is

$$X^{-1}(E) := \{\omega \in \Omega : X(\omega) \in E\} \subseteq \Omega.$$

To find this event, we collect the outcomes whose outputs fall in E. Whenever $X^{-1}(E)$ is an event, we define

$$\begin{aligned} P(X \in E) &:= P\big(X^{-1}(E)\big) \\ &= P\big(\{\omega \in \Omega : X(\omega) \in E\}\big). \end{aligned}$$

In particular, for a single value x,

$$P(X = x) := P\big(X^{-1}(\{x\})\big).$$

Here X is the random variable, while x is a particular real number.

Definition 3.5: Distribution of a random variable

The *distribution* of X is the rule that tells us with what probability X takes values in a given set E.

$$P_X(E) = P(X \in E).$$

Definition 3.6: Discrete random variable and PMF

A random variable X is *discrete* if there is a finite or countably infinite set[‡] $\mathcal{X} = \{x_1, x_2, \ldots\}$ such that $P(X \in \mathcal{X}) = 1$.

Its *probability mass function* (PMF) is

$$p_X(x) := P(X = x), \qquad x \in \mathbb{R}.$$

1. $p_X(x) \geq 0$ for every $x \in \mathbb{R}$.
2. The *support*, in the discrete sense used here, is $\text{supp}(X) := \{x \in \mathbb{R} : p_X(x) > 0\}$. Thus the support lists the values that have positive probability; values outside it have probability zero.
3. $\sum_{x \in \mathcal{X}} p_X(x) = 1$.

Definition 3.7: Continuous random variable and PDF

A random variable X is *continuous* (in the sense used here) if there exists a function f_X, called its *probability density function* (PDF), such that, for every $a \leq b$,

$$P(a \leq X \leq b) = \int_a^b f_X(x)\, dx.$$

1. $f_X(x) \geq 0$ for every $x \in \mathbb{R}$.
2. Following the convention in the slides, the *support* is $\text{supp}(X) := \{x \in \mathbb{R} : f_X(x) > 0\}$.
3. $\int_{\mathbb{R}} f_X(x)\, dx = 1$, since the total area under f_X equals $P(X \in \mathbb{R})$, and X certainly takes some real value.

Definition 3.8: Cumulative distribution function

Given a random variable X, its *cumulative distribution function* (CDF) is the function $F_X : \mathbb{R} \to [0, 1]$ defined by

$$F_X(x) := P(X \leq x) = P\big(X \in (-\infty, x]\big), \qquad x \in \mathbb{R}.$$

It gives the probability accumulated up to and including the threshold x. It is defined for both discrete and continuous random variables.

[‡]A finite set has finitely many elements. A set A is countably infinite if there exists a bijection $f : \mathbb{N} \to A$.

Proposition 3.1: Properties of the CDF (I)

A function $F(\cdot)$ mapping the real line to $[0,1]$ is a CDF for some probability measure $P(\cdot)$ if and only if:

1. $F(\cdot)$ is *non-decreasing*: $x_1 < x_2$ implies that $F(x_1) \leq F(x_2)$.
2. $F(\cdot)$ is *normalized*:
$$\lim_{x \to -\infty} F(x) = 0, \qquad \lim_{x \to +\infty} F(x) = 1.$$
3. $F(\cdot)$ is *right-continuous*: for all x we have
$$F(x) = F(x^+), \qquad F(x^+) := \lim_{t \downarrow x} F(t).$$

Proposition 3.2: Properties of the CDF (II/a)

Let $F_X(\cdot)$ be the CDF of a random variable X. Then:

1. For $x_1 < x_2$,
$$P(x_1 < X \leq x_2) = P\big(X \in (x_1, x_2]\big) = F_X(x_2) - F_X(x_1).$$
In the discrete case, be careful with the strict and non-strict inequalities.
2.
$$P(X > x) = P\big(X \in (x, +\infty)\big) = 1 - F_X(x).$$
3. If X is continuous with PDF $f_X(\cdot)$, then
$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt.$$
A PDF can be recovered from the CDF as
$$f_X(x) = F_X'(x)$$
at points where F_X is differentiable.

Proposition 3.3: Properties of the CDF (II/b)

Let $F_X(\cdot)$ be the CDF of a random variable X. Then:

1. $$P(x_1 < X \le x_2) = P\big(X \in (x_1, x_2]\big) = F_X(x_2) - F_X(x_1), \qquad x_1 < x_2.$$
2. $$P(X > x) = P\big(X \in (x, +\infty)\big) = 1 - F_X(x).$$
3. If X is discrete with PMF $p_X(\cdot)$ and support $\mathcal{X} = \{x_1, x_2, \ldots\}$, then
$$F_X(x) = \sum_{x_i \le x} p_X(x_i) = P(X < x) + p_X(x).$$
Hence
$$p_X(x) = F_X(x) - F_X(x^-), \qquad F_X(x^-) := \lim_{t \uparrow x} F_X(t).$$

Definition 3.9: Quantile function: continuous case

Let X be a continuous random variable with CDF F_X. For a probability level $0 < p < 1$, its *quantile function* is
$$Q(p) := \inf\{x \in \mathbb{R} : F_X(x) = p\}.$$
It gives the smallest value x such that $F_X(x) = P(X \le x) = p$. We also write q_p instead of $Q(p)$.

Definition 3.10: Quantile function: general case

Let X be any random variable, discrete or continuous, with CDF F_X. For a probability level $0 < p < 1$, its *quantile function* is
$$Q(p) := \inf\{x \in \mathbb{R} : F_X(x) \ge p\}.$$
It gives the smallest threshold x such that $P(X \le x)$ is at least p. We also write q_p instead of $Q(p)$.

Definition 3.11: Uniform distribution

A continuous random variable X has a *uniform distribution* with parameters $a < b$, where $a, b \in \mathbb{R}$, if its PDF and CDF are
$$f_X(x) = \frac{1}{b-a} \mathbf{1}_{[a,b]}(x),$$
$$F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x > b. \end{cases}$$
We denote this distribution by $\mathrm{Unif}(a, b)$ and write $X \sim \mathrm{Unif}(a, b)$. Here Unif abbreviates *uniform*, and $\sim$ means *has distribution*.

Definition 3.12: Exponential distribution

We write $T \sim \text{Exp}(\lambda)$, with rate $\lambda > 0$, when its PDF and CDF are

$$f_T(t) = \lambda e^{-\lambda t} \mathbf{1}_{[0,+\infty)}(t), \qquad F_T(t) = \begin{cases} 0, & t < 0, \\ 1 - e^{-\lambda t}, & t \geq 0. \end{cases}$$

Definition 3.13: Bernoulli distribution

For $0 \leq p \leq 1$, we write $X \sim \text{Ber}(p)$ when

$$p_X(x) = \begin{cases} 1 - p, & x = 0, \\ p, & x = 1, \\ 0, & \text{otherwise.} \end{cases}$$

The parameter p is the success probability. The two masses add to 1.

Definition 3.14: Binomial distribution

For an integer $n \geq 1$ and $0 < p < 1$, $X \sim \text{Bin}(n, p)$ has PMF

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n,$$

and zero elsewhere. For $p = 0$, $X = 0$ with probability 1; for $p = 1$, $X = n$ with probability 1.

Definition 3.15: Beta distribution

For $\alpha, \beta > 0$, the density of $P \sim \text{Beta}(\alpha, \beta)$ is

$$\pi(p) = \frac{p^{\alpha-1}(1-p)^{\beta-1}}{B(\alpha, \beta)}, \qquad 0 < p < 1,$$

and it is zero elsewhere. The denominator is the normalizing constant

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}\, dt.$$

Definition 3.16: Gamma distribution

Using the Gamma function[§] defined above, we write $T \sim \text{Gamma}(a, \lambda)$ when its PDF is

$$f_T(t) = \frac{\lambda^a}{\Gamma(a)} t^{a-1} e^{-\lambda t} \mathbf{1}_{(0,+\infty)}(t).$$

[§]In the notes, $\Gamma(a) = \int_0^{+\infty} t^{a-1} e^{-t}\, dt$ for $a > 0$.

Chapter 4. Random vectors

Definition 4.1: Random vector

On a common probability space $(\Omega, \mathcal{F}, P)$, a **random vector** is a vector of random variables

$$\underline{X} : \Omega \longrightarrow \mathbb{R}^p, \qquad \underline{X}(\omega) = \begin{pmatrix} X_1(\omega) \\ \vdots \\ X_p(\omega) \end{pmatrix}.$$

The integer p is the dimension of the vector. Each component $X_j : \Omega \to \mathbb{R}$ is a random variable.

Definition 4.2: Independence of random variables

Two random variables X and Y are independent when

$$P(X \in I,\ Y \in J) = P(X \in I)P(Y \in J) \quad \text{for every pair of intervals } I, J.$$

Definition 4.3: Joint probability mass function

The **joint PMF** assigns a probability to each pair of values[¶]:

$$p_{X,Y}(x, y) = P(X = x, Y = y) = P\big(\{X = x\} \cap \{Y = y\}\big).$$

It satisfies

$$p_{X,Y}(x, y) \geq 0, \qquad \sum_{x \in S_X} \sum_{y \in S_Y} p_{X,Y}(x, y) = 1.$$

Some pairs in $S_X \times S_Y$ may have probability zero. The joint PMF completely determines the distribution of (X, Y).

Proposition 4.1: Marginal PMFs

The marginal distributions are obtained by summing out the other variable:

$$p_X(x) = \sum_{y \in S_Y} p_{X,Y}(x, y), \qquad p_Y(y) = \sum_{x \in S_X} p_{X,Y}(x, y).$$

[¶]Here X and Y are discrete, with supports S_X and S_Y.

Definition 4.4: Conditional PMFs

For a value y with $p_Y(y) > 0$,

$$p_{X|Y}(x \mid y) = P(X = x \mid Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Similarly, for x with $p_X(x) > 0$,

$$p_{Y|X}(y \mid x) = P(Y = y \mid X = x) = \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

For each fixed conditioning value, the conditional probabilities are nonnegative and sum to 1 over the remaining variable.

Definition 4.5: Independence of discrete random variables

Two discrete random variables X and Y are independent if and only if

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \qquad \text{for every } (x, y) \in S_X \times S_Y.$$

Equivalently,

$$p_{X|Y}(x \mid y) = p_X(x) \quad \text{for every } x \text{ and every } y \text{ with } p_Y(y) > 0.$$

Definition 4.6: Joint cumulative distribution function

For any pair of random variables, discrete or otherwise,

$$F_{X,Y}(x, y) = P(X \le x, Y \le y), \qquad (x, y) \in \mathbb{R}^2.$$

This is the probability of the lower-left region $(-\infty, x] \times (-\infty, y]$ in the plane.

Proposition 4.2: Marginal CDFs

The marginal CDFs are obtained as limits of the joint CDF:

$$F_X(x) = \lim_{y \to +\infty} F_{X,Y}(x, y), \qquad F_Y(y) = \lim_{x \to +\infty} F_{X,Y}(x, y).$$

Proposition 4.3: X, Y independent $\iff$ joint CDF $= F_X F_Y$

Let X and Y be **arbitrary random variables**, with CDFs F_X and F_Y. They are **independent if and only if** their joint CDF satisfies

$$F_{X,Y}(x, y) = F_X(x)F_Y(y) \qquad \text{for all } x, y \in \mathbb{R}.$$

Definition 4.7: Joint PDF

A **joint probability density function** $f_{\underline{X}} : \mathbb{R}^p \to [0, +\infty)$ satisfies

$$\int_{\mathbb{R}^p} f_{\underline{X}}(\underline{x})\, d\underline{x} = 1,$$

and gives the probability of a measurable region $A \subseteq \mathbb{R}^p$ through

$$P(\underline{X} \in A) = \int_A f_{\underline{X}}(\underline{x})\, d\underline{x}.$$

Definition 4.8: Kernel of a PDF

A **kernel of a PDF** is a nonnegative function $g(\underline{x})$ proportional to that PDF. It specifies the shape of the density, leaving a positive multiplicative constant to be determined.

To obtain the PDF, compute the total integral

$$c = \int_{\mathbb{R}^p} g(\underline{x})\, d\underline{x}.$$

If $0 < c < +\infty$, then

$$f_{\underline{X}}(\underline{x}) = \frac{g(\underline{x})}{c}$$

is the corresponding PDF. The constant c is the **normalizing constant**.

Proposition 4.4: Marginal PDFs

Let $\underline{X} = (X_1, \ldots, X_p)^\mathsf{T}$ have joint PDF $f_{\underline{X}}$. The density of any subset of components is obtained by integrating out the remaining ones. Reordering the components if necessary, the first $k < p$ components have density

$$f_{X_1,\ldots,X_k}(x_1, \ldots, x_k) = \int_{\mathbb{R}^{p-k}} f_{\underline{X}}(x_1, \ldots, x_p)\, dx_{k+1} \cdots dx_p.$$

In particular, for $p = 2$ with components (X, Y),

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x, y)\, dy, \qquad f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y)\, dx.$$

Definition 4.9: Conditional PDFs

When a joint PDF exists, a conditional density of X given $Y = y$ can be defined, for $f_Y(y) > 0$, by

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

Similarly, for $f_X(x) > 0$,

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

The conditioning value is held fixed. Integrating the conditional density over the other variable gives 1.

Proposition 4.5: X, Y independent $\iff$ joint PDF $= f_X f_Y$

Suppose X and Y **each have a PDF**, denoted by f_X and f_Y. They are **independent if and only if** their joint PDF exists and satisfies

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) \qquad \text{almost everywhere.}$$

Definition 4.10: Mixed joint function

Let $X \mid (P = p)$ have PMF $b(k \mid p)$ on $\{0, \ldots, n\}$, and let P have prior PDF $\pi(p)$ on $(0, 1)$. The **mixed joint function** of (X, P) is

$$q(k, p) = b(k \mid p)\, \pi(p), \qquad k = 0, \ldots, n, \quad 0 < p < 1.$$

It gives the probability that X equals k and, at the same time, P falls in an interval (a, b):

$$\Pr(X = k,\ a < P < b) = \int_a^b q(k, p)\, dp, \qquad 0 \le a < b \le 1.$$

Proposition 4.6: Marginal PMF in the mixed model

The marginal PMF of X, denoted by $m(k)$, is

$$m(k) = \Pr(X = k) = \int_0^1 b(k \mid p)\, \pi(p)\, dp, \qquad k = 0, \ldots, n.$$

Proposition 4.7: Posterior density in the mixed model

Let k be a count with $m(k) > 0$. Given $X = k$, the latent variable P has the **posterior density**

$$f_{P|X}(p \mid k) = \frac{b(k \mid p)\, \pi(p)}{m(k)}, \qquad 0 < p < 1.$$

Definition 4.11: Bivariate copula

A **copula** is a joint CDF of two random variables with standard uniform marginals:

$$C(u, v) = P(U \le u, V \le v), \qquad U, V \sim \text{Unif}(0, 1).$$

The variables U and V need not be independent. On $[0, 1]^2$, the marginal conditions are $C(u, 1) = u$ and $C(1, v) = v$.

Theorem 4.1: Sklar's theorem: bivariate form

For any joint CDF $F_{X,Y}$ with marginals F_X and F_Y, there exists a copula C such that

$$F_{X,Y}(x,y) = C\big(F_X(x), F_Y(y)\big).$$

If both marginals are continuous, the copula is unique. Otherwise, it is uniquely determined only on $\mathrm{Ran}(F_X) \times \mathrm{Ran}(F_Y)$, the product of the sets of values taken by the marginal CDFs.

Conversely, combining any two marginal CDFs with a copula using this formula produces a joint CDF with those marginals.

Definition 4.12: Independent and identically distributed sample

A random sample $(X_1, \ldots, X_n)$ is **IID** from F_X if

- the random variables $X_1, \ldots, X_n$ are mutually independent;
- each X_i has the same CDF F_X.

Here n is the sample size. Uppercase X_i denotes a random observation before sampling; lowercase x_i denotes its observed value. The realized dataset is $\mathcal{D}_n = (x_1, \ldots, x_n)$.

Definition 4.13: Likelihood

For fixed observed data $\mathcal{D}_n = (x_1, \ldots, x_n)$ and a model parameter $\theta \in \Theta$, the **IID likelihood** is

$$L(\theta; x_1, \ldots, x_n) = \begin{cases} \displaystyle\prod_{i=1}^{n} p_\theta(x_i), & \text{discrete model,} \\ \displaystyle\prod_{i=1}^{n} f_\theta(x_i), & \text{model with a PDF.} \end{cases}$$

Definition 4.14: Maximum likelihood estimation

A **maximum likelihood estimator (MLE)** chooses

$$\widehat{\theta}_n \in \arg\max_{\theta \in \Theta} L(\theta; x_1, \ldots, x_n),$$

when a maximizer exists. Here $\widehat{\theta}_n$ is the parameter estimate calculated from the n observations, and arg max denotes the set of parameter values attaining the largest likelihood.

Definition 4.15: Empirical cumulative distribution function

Given observations $x_1, \ldots, x_n$, the **empirical CDF** is

$$\widehat{F}_n(t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{x_i \le t\}}.$$

It is the fraction of observations at or below t. The corresponding empirical distribution assigns mass $1/n$ to each observation, counting repetitions.

Chapter 5. Stochastic simulations

Definition 5.1: Pseudorandom number generator

In a simplified description, a *pseudorandom number generator* (PRNG) uses a deterministic function

$$f : (0,1) \longrightarrow (0,1)$$

and a starting value, or *seed*, $u_0 \in (0,1)$ to generate

$$u_0, \quad u_1 = f(u_0), \quad u_2 = f(f(u_0)), \quad \ldots, \quad u_n = f^n(u_0).$$

The function is chosen so that the sequence imitates the statistical behaviour of an IID Uniform$(0,1)$ sample. Here $f^n(u_0)$ means applying f repeatedly n times.

Definition 5.2: Linear congruential generator

A *linear congruential generator* (LCG) uses integer parameters α, β, and $M > 1$, and an integer seed $x_0 \in \{0, 1, \ldots, M-1\}$. Its update rule is

$$x_{k+1} = f(x_k) = (\alpha + \beta x_k) \bmod M.$$

Here $\bmod M$ gives the remainder after division by M, so every state x_k belongs to $\{0, 1, \ldots, M-1\}$.

Proposition 5.1: Quantile transform

Let F be any CDF and let

$$Q(p) = \inf\{x \in \mathbb{R} : F(x) \geq p\}, \qquad 0 < p < 1,$$

be its quantile function. If $U \sim \text{Unif}(0,1)$ and $Y = Q(U)$, then Y has CDF F:

$$F_Y(y) = P(Y \leq y) = F(y), \qquad y \in \mathbb{R}.$$

Chapter 6. Exploratory data analysis

Definition 6.1: Class of models

In this density-estimation problem, a **class of models** $\mathcal{F}$ is a set of candidate PDFs. Each $f \in \mathcal{F}$ represents one possible population distribution.

Definition 6.2: Penalized log-likelihood

For fixed observed data $\mathcal{D}_n = (x_1, \ldots, x_n)$ from an IID sample, consider a candidate PDF f. Let $J(f) \geq 0$ be its penalty, and let $\lambda > 0$ be the weight given to that penalty. The **penalized log-likelihood** is

$$\log L(f; \mathcal{D}_n) - \lambda J(f) = \sum_{i=1}^{n} \log f(x_i) - \lambda J(f).$$

Relations

From the PMF to the CDF

What we are trying to answer

If we know the probability of each individual value of X, how do we calculate the probability that X is at most x?

For a discrete random variable, the event $X \leq x$ includes every possible value k that is at most x. These events $\{X = k\}$ are pairwise disjoint, so we add their probabilities:

$$\underbrace{F_X(x)}_{\text{CDF: to find}} = P(X \leq x) = \sum_{\substack{k \in \text{supp}(X) \\ k \leq x}} \underbrace{p_X(k)}_{\text{PMF: known}}, \qquad x \in \mathbb{R}.$$

From the PDF to the CDF

What we are trying to answer

We know how probability is spread over the possible values of X (its PDF). How do we calculate the probability of getting a value less than or equal to x?

For a continuous random variable with PDF f_X, we calculate $P(X \leq x)$ by integrating the PDF over $(-\infty, x]$:

$$\underbrace{F_X(x)}_{\text{CDF: to find}} = P(X \leq x) = \int_{-\infty}^{x} \underbrace{f_X(t)}_{\text{PDF: known}} \, dt, \qquad x \in \mathbb{R}.$$

From the CDF to the PMF and PDF

What we are trying to answer

We know the CDF F_X. How do we recover the PMF p_X if X is discrete, or the PDF f_X if X is continuous?

What we know. The CDF is defined by

$$F_X(x) := P(X \leq x).$$

For every threshold x, it gives the probability of obtaining a value less than or equal to x. We want to recover the PMF or the PDF, depending on the type of X.

Discrete case: recover the PMF. The PMF is defined by

$$p_X(x) := P(X = x).$$

The CDF includes all values up to x. To isolate the probability of the single value x, we subtract the probability of values strictly less than x:

$$P(X = x) = P(X \leq x) - P(X < x).$$

Since the left limit satisfies

$$F_X(x^-) := \lim_{t \uparrow x} F_X(t) = P(X < x),$$

we obtain

$$\boxed{\underbrace{p_X(x)}_{\text{PMF: to find}} = \underbrace{F_X(x) - F_X(x^-)}_{\text{CDF: known}}.}$$

Thus the PMF at x is the size of the jump of the CDF at x.

Continuous case: recover the PDF. A PDF is a non-negative function f_X such that

$$P(a \leq X \leq b) = \int_a^b f_X(t)\,dt, \qquad \int_{\mathbb{R}} f_X(t)\,dt = 1.$$

In particular,

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt.$$

The CDF is therefore the integral of the PDF. To recover a PDF, we differentiate the CDF:

$$\boxed{\underbrace{f_X(x)}_{\text{PDF: to find}} = \frac{d}{dx} \underbrace{F_X(x)}_{\text{CDF: known}}.}$$

This recovers a PDF at points where the derivative exists, by the fundamental theorem of calculus.